DEEPANSHU MEHTA

Senior Software Engineer • Backend & Full-Stack • GenAI / LLM Systems

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Open to Relocation

PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer with 6+ years of experience designing and delivering scalable backend systems, full-stack applications, and AI-powered solutions. Deep expertise in Python (Django, FastAPI), cloud platforms (AWS, Azure), and modern LLM/GenAI frameworks (LangChain, LangGraph). Proven track record across healthcare, EdTech, e-commerce, and enterprise domains. Experienced in agentic AI systems, RAG pipelines, event-driven architectures, and DevOps practices including containerisation and CI/CD. Actively seeking to leverage distributed systems and AI expertise at Google.

TECHNICAL SKILLS

Languages:	Python (6 yrs), JavaScript / TypeScript (4 yrs), SQL, Bash
Backend:	FastAPI, Django, Flask, REST APIs, gRPC, Async / Concurrent Programming, Multithreading
Frontend:	ReactJS, Angular, TypeScript, HTML5, CSS3, pnpm
GenAI / ML:	LangChain, LangGraph, RAG Pipelines, Agentic AI, OpenAI / Anthropic APIs, Prompt Engineering, TensorFlow, PyTorch, scikit-learn, NLP
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Snowflake, Elasticsearch
Cloud & DevOps:	AWS (Lambda, SQS, EventBridge, KMS, IAM, S3), Azure (Key Vault, Functions), Docker, Kubernetes, Databricks, PySpark, JFrog, CI/CD
Architecture:	Microservices, Event-Driven (SQS, RabbitMQ), Serverless, SOLID / CLEAN Principles, TDD (PyTest), Connection Pooling, LDAP / OpenID / JWT Auth

PROFESSIONAL EXPERIENCE

Consultant | Deloitte USI — Nov 2024 – Present

Domain: Healthcare • Stack: Python, FastAPI, ReactJS, pnpm, LangChain, LangGraph, Snowflake, Redis, Azure, Databricks, JFrog

- Architected a RAG-based document intelligence platform for healthcare data governance: users upload clinical/policy documents as context, and the system auto-generates a glossary of table/column descriptions from metadata schemas, reducing manual documentation effort by ~70%.
- Owned frontend feature development for enterprise web applications using ReactJS with pnpm workspace setups, enhancing component performance and stream-lining package dependency management.
- Built an agentic AI pipeline using LangGraph to autonomously crawl and compare competitor healthcare policies against client policies, identify regulatory loopholes, and auto-draft hardened policy recommendations — currently in active production at Deloitte.
- Partnered closely with the DevOps team to streamline build pipelines, managing secure application artifacts and package distribution using JFrog Artifactory.
- Designed a high-throughput FastAPI wrapper with Elasticsearch: implemented elastic connection pooling and multithreading to handle thousands of concurrent healthcare data requests with zero message loss.
- Established Redis as Celery messaging Queue for handling the Async Tasks.

Senior Software Engineer | Quadrant Technologies (Deputed: Deloitte) — Nov 2023 – Nov 2024

Domain: Healthcare • Stack: Python, FastAPI, LangChain, LangGraph, Snowflake, Redis, MongoDB, Azure, Databricks, ReactJS

- Designed a high-throughput FastAPI wrapper with Multithreading & Multiprocessing: implemented elastic connection pooling and multithreading to handle thousands of concurrent healthcare data requests with zero message loss.
- Established Redis caching layer with multithreaded cache population for frequently accessed healthcare records, reducing average API response time by 60% under peak load.
- Implemented enterprise-grade authentication via LDAP and OpenID Connect for HIPAA-aligned access control; integrated Azure Key Vault for zero-plaintext secret management.
- Built async background task framework compatible with microservice architecture for healthcare data pipelines; integrated structured logging and distributed error-handling traceable across services.
- Optimised database and Redis connection pools on Snowflake and PostgreSQL, improving throughput by 40% for high-concurrency healthcare analytics workloads.

Senior Software Developer | CrewScale (Deputed: Everest Fleet) — Mar 2023 – Nov 2023

Stack: Python, Django, MySQL, AWS Lambda, SQS, EventBridge, ReactJS

- Led migration of monolithic services to serverless architecture on AWS Lambda, reducing infrastructure costs by 35%.
- Designed event-driven queue system using AWS SQS with concurrent Lambda triggers via EventBridge, improving order processing throughput by 50%.
- Restructured legacy relational database schema for Everest Fleet's fleet management platform, achieving 30% reduction in query execution time.
- Built an in-house analytics module providing real-time fleet KPIs and driver performance dashboards for ops teams.

Senior Software Developer | Instahyre — Oct 2022 – Jan 2023

Stack: Python, Django, PostgreSQL, MySQL, TypeScript, Angular, Elasticsearch, RabbitMQ

- Contributed to Instahyre's hiring portal as a full-stack engineer (ReactJS + Django), focusing on recruiter-side pipeline management and candidate tracking features.
- Implemented idempotent priority queues with RabbitMQ for reliable background task execution, eliminating duplicate processing bugs.
- Optimised Elasticsearch indexing and query pipeline, reducing search latency by 45% across job and candidate indices.
- Resolved critical frontend performance regressions in AngularJS, improving page load times and overall recruiter UX.

Software Developer | Cashify — Aug 2021 – Sep 2022

Stack: Python, Django, FastAPI, PostgreSQL, MySQL, JavaScript

- Developed and maintained Cashify's internal Inventory Management System, handling SKU lifecycle from acquisition to resale.
- Applied SOLID and Clean Architecture principles to refactor legacy codebase using Black formatter, increasing code maintainability score significantly.
- Converted synchronous Django modules to async, improving I/O-bound API response times by 35%.
- Adopted TDD methodology with PyTest, increasing test coverage from ~20% to 80%+ across critical inventory modules.
- Built Django Admin panels for business operations teams, enabling self-service data management and reducing ops ticket volume.

Junior Software Developer | Nickelfox Technologies — Jun 2020 – Aug 2021

Stack: Python, Django, PostgreSQL, MySQL, Redis, JWT, AWS SES, FCM / APNs

- Built backend for a healthcare app monitoring user heartbeats and sleep patterns, integrating wearable data to generate doctor-facing health reports.
- Engineered a mood-based multimedia recommendation engine using collaborative filtering, improving user engagement by 40%.
- Developed a QR-code-driven eLearning backend: scanning a book's QR code triggers contextual video content on the mobile app.
- Implemented multi-channel notification service using APNs, AWS SES, and Firebase FCM for push, email, and in-app notifications.
- Designed JWT-based authentication and role-based authorisation system across all platform services.
- Refactored existing services from MVC to MVCS architecture, reducing average response time by 25%.

KEY PROJECTS & GENAI WORK

Agentic Policy Analysis System — LangGraph • LangChain • FastAPI • Redis • Databricks

- Designed a multi-agent LangGraph pipeline that autonomously discovers competitor insurance/financial policies from the web, performs semantic comparison against client documents, identifies regulatory loopholes, and produces hardened policy recommendations — cutting manual policy review cycles from days to hours.

RAG Document Intelligence & Glossary Generator — FastAPI • LangChain • Snowflake • Azure

- Built a production RAG system enabling users to upload domain documents as context; the pipeline parses metadata tables (table name, column name) and uses LLM inference to auto-generate human-readable table and column descriptions, dramatically improving data governance documentation.

High-Throughput Elasticsearch API Gateway — FastAPI • Elasticsearch • Multithreading • Redis

- Implemented a fault-tolerant FastAPI gateway around Elasticsearch, leveraging connection pooling, thread-safe request queuing, and Redis-backed response caching — achieving zero message loss under stress loads that previously caused system failures.

EDUCATION

B.Tech – Electronics & Communication Engineering — Aug 2016 – Jun 2020

Bennett University, Greater Noida, Uttar Pradesh, India

CERTIFICATIONS & TRAINING

Artificial Intelligence with Python — CDAC — Jul 2020

Covered ML/DL fundamentals, CNNs, image classification, NLP basics; hands-on with TensorFlow, PyTorch.

MongoDB NoSQL Basics — MongoDB University — Aug 2022

CRUD operations, aggregation, indexing, data modeling, transactions; built a project in PyMongo.